

# ALGEBRA BASICS



GRADE 6

# ***What is Substitution?***

Substitution is an algebraic technique used to solve equations and simplify expressions by replacing one or more variables (letters) with specific numbers.



***It's like 'subbing' a  
player in a soccer or  
basketball match!***

# ***How do we substitute?***

Step 1: Understand Variables

Step 2: Identify the Variable/s to Substitute

Step 3: Substitute and Simplify

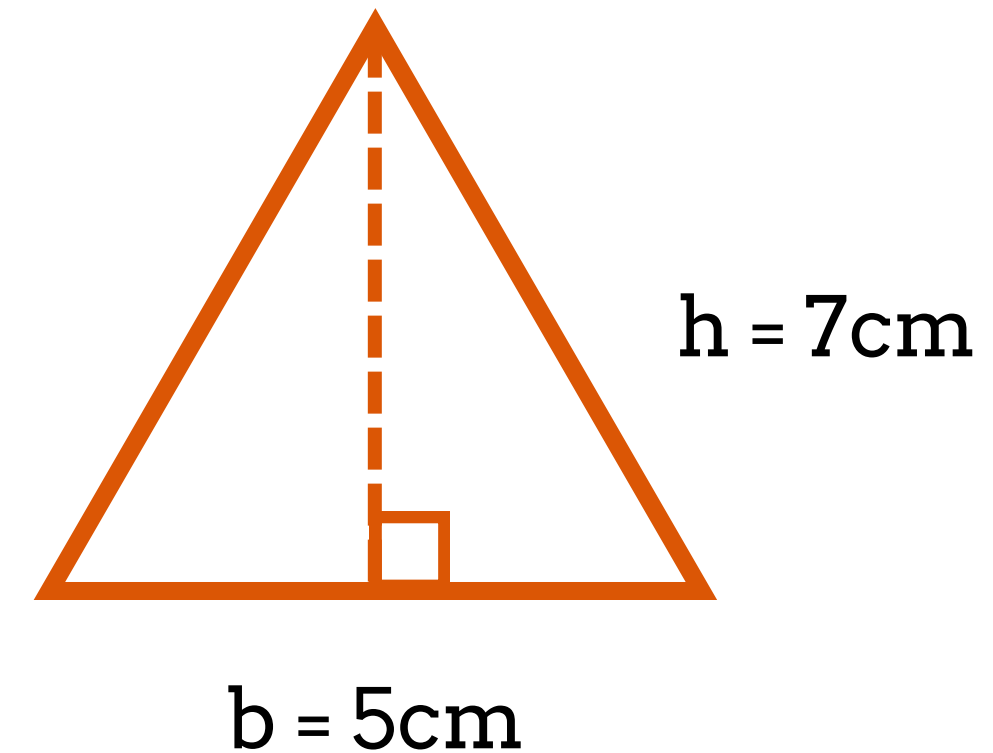


# ***Let's Try***

The formula for the area of a triangle is given by:  **$a = bh/2$**

## **Step 1: Understand Variables**

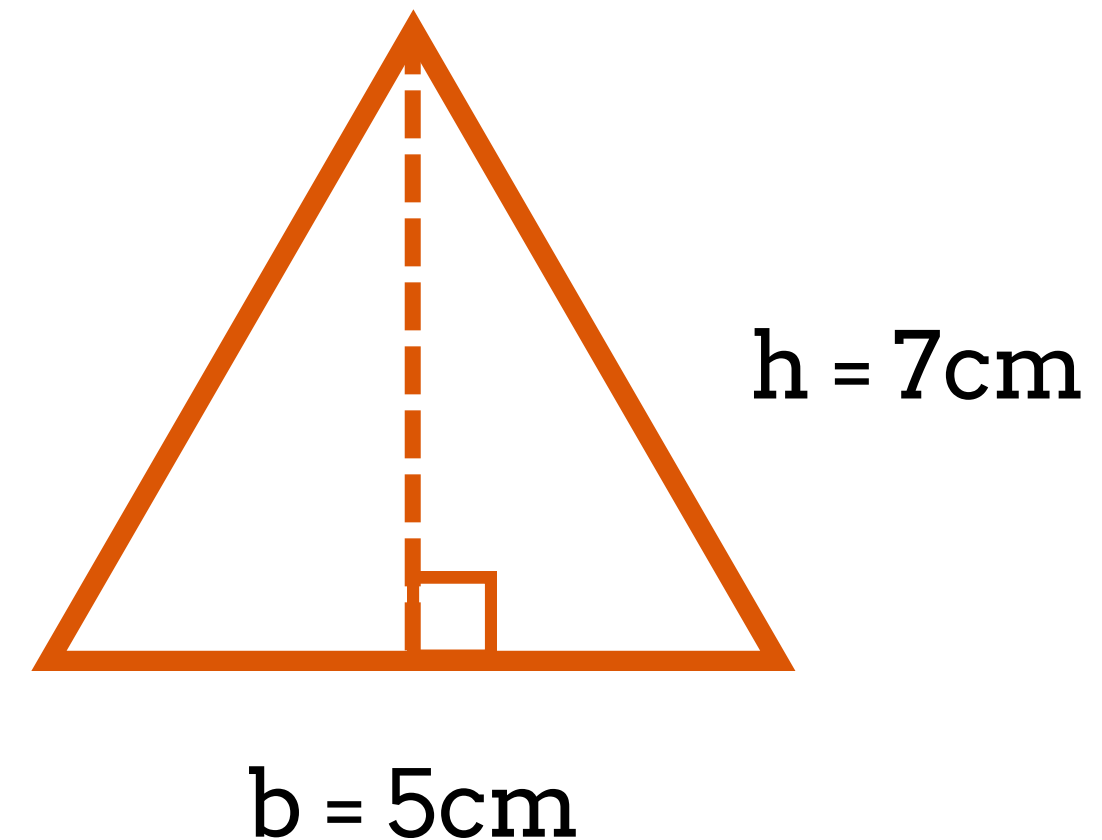
- $a$  = area of the triangle
- $b$  = base length of the triangle
- $h$  = height of the triangle



## Step 2: Identify the Variable to Substitute

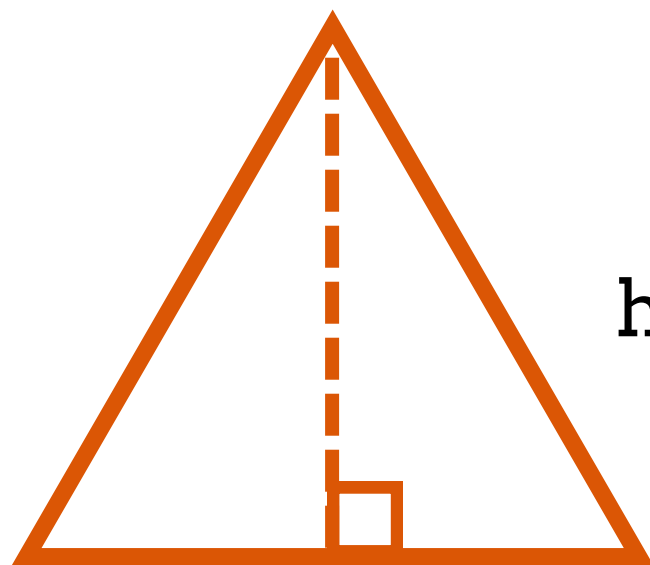
We can substitute values into  $b$  and  $h$  because these values have been given to us in the diagram.

We can use these values to find the area of the triangle (a).



### Step 3: Substitute and Simplify

Substitute the given values into b and h to simplify and solve for a.



$h = 7\text{cm}$

$b = 5\text{cm}$

$a = \text{unknown}$

$h = 7\text{cm}$

$b = 5\text{cm}$

$a = bh/2$

$a = 5 \times 7 / 2$

$a = 35 / 2$

$a = 17.5\text{cm}^2$

**The area of the triangle is  $17.5\text{cm}^2$ .**



***Here's another***

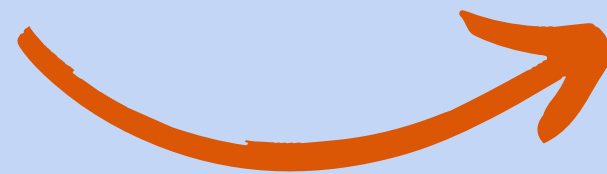
The formula for simple interest is given by:  **$I = PRT$**

### **Step 1: Understand Variables**

- $I$  = amount of interest (\$)
- $P$  = initial investment (\$)
- $R$  = interest rate (as a decimal)
- $T$  = time (years)

## Step 2: Identify the Variable/s to Substitute

If we are investing \$500 in the bank for 3 years at an interest rate of 3.5%, how much interest will we receive?



$I = \text{unknown}$

$P = 500$

$R = 0.035$

$T = 3$

### Step 3: Substitute and Simplify

$I = \text{unknown}$

$P = 500$

$R = 3.5$

$T = 3$



$I = PRT$

$I = 500 \times 0.035 \times 3$

$I = 52.5$

**We would receive \$52.50 in interest.**

## QUESTION 1

## LINEAR ROUND

What is the coefficient in the expression  
 $6x^2 + 4$ ?

A

6

B

2

C

x

D

4

## QUESTION 1

## LINEAR ROUND

What is the coefficient in the expression  
 $6x^2 + 4$ ?

**ANSWER:**

**A**

**6**

## QUESTION 2

## LINEAR ROUND

In the expression  $4x + 6 - 2x$ , which terms are like terms?

A

$4x$  and  $6$

B

$4x$  and  $-2x$

C

$6$  and  $-2x$

D

None

## QUESTION 2

## LINEAR ROUND

In the expression  $4x + 6 - 2x$ , which terms are like terms?

**ANSWER:**

**B**

$4x$  and  $-2x$

### QUESTION 3

### LINEAR ROUND

Which of the following is a term in the expression  $3y + 8 - 4y$ ?

A

$y$

B

3

C

$-4$

D

8

### QUESTION 3

### LINEAR ROUND

Which of the following is a term in the expression  $3y + 8 - 4y$ ?

**ANSWER:**

**D**

8

## QUESTION 4

## LINEAR ROUND

Simplify the following expression:

$$4(x-2) + 5$$

A

$$4x + 7$$

B

$$4x - 7$$

C

$$4x - 3$$

D

$$4x + 3$$

## QUESTION 4

## LINEAR ROUND

Simplify the following expression:

$$4(x-2) + 5$$

**ANSWER:**

**C**

$$4x - 3$$

## QUESTION 5

## LINEAR ROUND

How many terms are in the expression  $3(x + 2) - 4x + 1$  after it is simplified?

A

1

B

2

C

3

D

4

## QUESTION 5

## LINEAR ROUND

How many terms are in the expression  
 $3(x + 2) - 4x + 1$  after it is simplified?

**ANSWER:**

**B**

**2**

## QUESTION 6

## SQUARE ROUND

Find the value of x:

$$3x + 2 = 11$$

A

2

B

3

C

4

D

5

## QUESTION 6

## SQUARE ROUND

Find the value of x:

$$3x + 2 = 11$$

**ANSWER:**

**B**

**3**

## QUESTION 7

## SQUARE ROUND

Which of the following equations does NOT have the solution  $x = 3$ ?

A

$$2x + 1 = 7$$

B

$$3x = 9$$

C

$$x - 4 = -1$$

D

$$x - 6 = 3$$

## QUESTION 7

## SQUARE ROUND

Which of the following equations does NOT have the solution  $x = 3$ ?

**ANSWER:**

**D**

$$x - 6 = 3$$

## QUESTION 8

## SQUARE ROUND

Which of the following equations has NO solution?

A

$$4X = X + 3X$$

B

$$2X + 4 = 10$$

C

$$X - 2 = 5$$

D

$$3X - 5 = 3X + 2$$

## QUESTION 8

## SQUARE ROUND

Which of the following equations has NO solution?

**ANSWER:**

**D**  $3X - 5 = 3X + 2$

## QUESTION 1

## CUBE ROUND

Jamie has 3 more than twice the number of pens that Leah has.

If Leah has  $x$  pens, write the expression that shows how many pens Jamie has.

## QUESTION 1

## CUBE ROUND

Jamie has 3 more than twice the number of pens that Leah has. If Leah has  $x$  pens, write the expression that shows how many pens Jamie has.

**ANSWER:**

$$3 + 2x$$

## QUESTION 2

## CUBE ROUND

Solve the equation:  
$$2(x - 4) + 3 = 13$$

## QUESTION 2

## CUBE ROUND

**ANSWER:**

$$2x - 8 + 3 = 13$$

$$2x - 5 = 13$$

$$2x = 18$$

$$x = 9$$

### QUESTION 3

### CUBE ROUND

Liam is 4 years less than twice Emma's age.

If Emma is  $x$  years old, write the expression that shows how old Liam is.

### QUESTION 3

### CUBE ROUND

Liam is 4 years less than twice Emma's age. If Emma is  $x$  years old, write the expression that shows how old Liam is.

**ANSWER:**

$$2x - 4$$

## QUESTION 4

## CUBE ROUND

A phone plan charges a \$15 monthly fee plus \$2 for every gigabyte of data used.

If  $x$  is the number of gigabytes, write the expression that shows the total cost of the plan.

## QUESTION 4

## CUBE ROUND

A phone plan charges a \$15 monthly fee plus \$2 for every gigabyte of data used.  
If  $x$  is the number of gigabytes, write the expression that shows the total cost of the plan.

**ANSWER:**

$$2x + 15$$



THANK YOU